

BAYAN ALSULTAN

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ABOUT ME

Data Science student with hands-on experience in Machine Learning, AI, Data Analytics, and NLP. Skilled in building predictive models, developing data-driven applications, and transforming complex datasets into actionable insights. Proficient in Python, SQL, Apache Spark, Power BI, and modern AI frameworks.

EDUCATION

Bachelor of Data Science

Saudi Electronic University

Expected Graduation: **June 2027**

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java

Machine Learning & AI: Machine Learning, NLP, Scikit-learn, AraBERT, Hugging Face, SHAP, LIME

Data Analysis & Visualization: Data Analysis, EDA, Predictive Modeling, Data Visualization, Pandas, NumPy

Big Data & BI: Apache Spark, Power BI

Tools & Platforms: Git, Excel

PROJECTS

AI-Powered Student Dropout Prediction & Early Warning System (Senior Project - Ongoing)

Technologies: Python, Machine Learning, SHAP, LIME, Data Analytics

- Developed predictive models to identify students at risk of dropout using academic and behavioral data.
- Evaluated and compared multiple machine learning algorithms including Logistic Regression, Random Forest, and Neural Networks.
- Implemented Explainable AI (XAI) techniques using SHAP and LIME to improve model transparency and interpretability.
- Designed an Early Warning System (EWS) that categorizes students into risk levels (Low, Medium, High).
- Created system architecture diagrams including ERD, Class Diagrams, and Sequence Diagrams.
- Contributed to the development of an interactive dashboard for monitoring student risk indicators and analytics.

Arabic Poetry Recommendation System Using AI

Technologies: Python, AraBERT, Hugging Face, NLP

- Developed an end-to-end NLP solution for Arabic emotion detection and poetry recommendation.
- Fine-tuned the AraBERT transformer model for multi-class emotion classification.
- Applied advanced preprocessing, tokenization, and data augmentation techniques to improve model performance.
- Achieved 82% classification accuracy, outperforming baseline deep learning models.
- Built an interactive web application for real-time emotion analysis and personalized poetry recommendations.

End-to-End Social Media Analytics System

Technologies: Python, Apache Spark, NLP, APIs, Power BI

- Developed an end-to-end social media analytics platform using Apache Spark and NLP to process large-scale Twitter and YouTube data.
- Applied text preprocessing and sentiment analysis techniques to extract audience perception and insights.
- Built Power BI dashboards to visualize sentiment trends, engagement metrics, and audience insights.

Ticket Sales Analytics Dashboard

Technologies: Power BI, Excel, Data Analytics

- Analyzed customer purchasing behavior and sales trends to support business decision-making.
- Performed data cleaning, validation, and transformation to ensure data quality.
- Conducted exploratory data analysis (EDA) to uncover actionable insights.
- Designed interactive Power BI dashboards to support business reporting and decision-making.

CERTIFICATIONS & PROGRAMS

Google Data Analytics Professional Certificate

March 2024

Apple UX/UI Foundation Certificate

November 2023

Samsung Innovation Campus

December 2025

- Training in Artificial Intelligence, Machine Learning, and real-world applied projects

LANGUAGES

Arabic

English